

Research part for the proposal of a SNSF Scientific Event

Title: Biodiversity Dynamics in Coevolutionary Meta-ecosystems (COMETA)

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Summary

Coevolution, the trait change driven by reciprocal selection within and among interacting species, has been invoked as a primary mechanism generating much of Earth's biodiversity. Two main theories have been proposed to explain the dynamics of coevolutionary diversification. The *Fisherian sexual selection theory*, which states that female and male traits can lead to sexual isolation and speciation, and the *Geographic mosaic theory of coevolution*, which suggests that reciprocal changes in traits determining species interaction produce coevolutionary diversification as interspecific interactions control reproductive isolation, genetic and phenotypic divergence, and speciation. Despite the appealing of such theories for explaining much of Earth's biodiversity, understanding how coevolution of intra- and inter-specific traits accounting for ecological interactions alter biodiversity dynamics across heterogeneous landscapes remain elusive. Such understanding is particularly important in ecosystems with strong coevolution in species- and interaction-rich ecological networks because the ongoing global change might trigger many species extinctions and the collapse of ecosystems. In this vein, joining theoretical and empirical evolutionary ecologists to develop theoretical models that incorporate the empirical patterns of reciprocal trait changes will be critical to improve our predictions on biodiversity dynamics in species- and interaction-rich ecological communities. Therefore, the aim of the workshop and the research visit is to connect trait-species-space interactions data to new process-based coevolutionary metaecosystem models. Such integration might facilitate the exploration of patterns of coexistence of co-evolved species, their role in ecosystem function, and the risk of collapse of co-evolved ecosystems across different spatio-temporal scales.